

GHOST Finance: Protocol Architecture

A Technical Walkthrough of the Core Mechanisms

Based on GHOST Finance Litepaper v1.0 — February 2026

Built on Midnight Network

This document walks through the GHOST Finance protocol using a series of diagrams, each introduced with a brief explanation. The diagrams are designed to be read sequentially: from the high-level architecture down to the specific mechanics of price discovery, risk stratification, reputation scoring, and tokenomics. GHOST Finance is deployed exclusively on **Midnight Network**, a data-protection blockchain built on Cardano that enables confidential smart contracts via zero-knowledge proofs — making it the natural home for a protocol where sealed bids and pseudonymous credit history are first-class requirements.

Contents

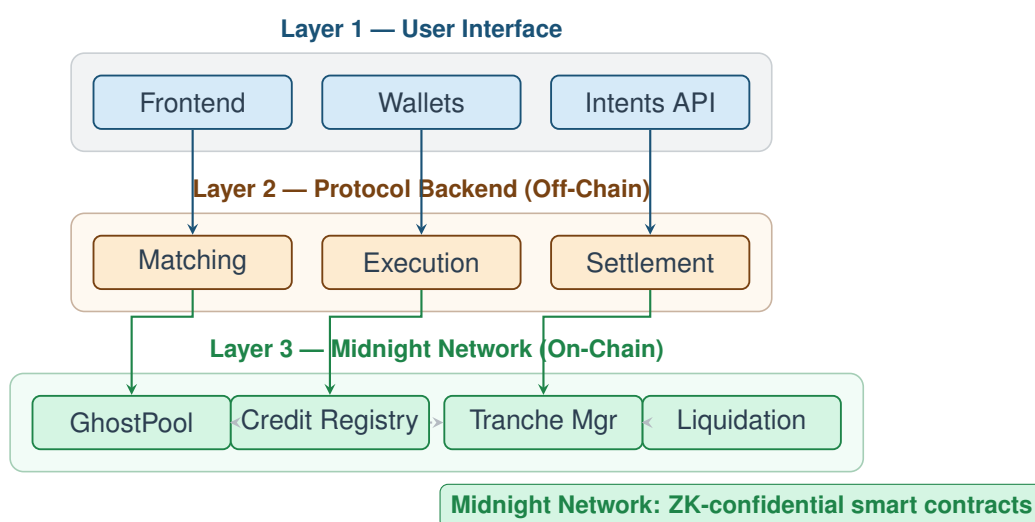
1	System Overview	2
2	Midnight Network: The Settlement Layer	3
3	Batch Auction Mechanism	4
3.1	The Problem with Continuous Markets	4
3.2	Intent Data Structures	4
3.3	Clearing Rate Calculation	5
4	Tranched Risk Architecture	6
4.1	Principal Split	6
4.2	Waterfall Distribution	6
5	Cross-Chain Deposit Flow	7
6	Haunt Score: On-Chain Credit Reputation	8
6.1	Motivation	8
6.2	Score Accumulation and Collateral Formula	8
6.3	The Five Manifestation Tiers	9
6.4	Score Lifecycle State Machine	9
7	GHOST Tokenomics	10
7.1	Token Distribution	10
7.2	Token Utility Map	10
8	Protocol Invariants and Security Model	11
9	End-to-End Transaction Flow	12

System Overview

GHOST Finance is a DeFi lending protocol built around three principles: **sealed-bid batch auctions** for price discovery, **tranching risk** for capital stratification, and **on-chain reputation** (the Haunt Score) for progressive capital efficiency. The protocol is deployed exclusively on **Midnight Network** — a privacy-preserving blockchain in the Cardano ecosystem that supports confidential smart contracts through zero-knowledge proofs. This choice is deliberate: Midnight's native data-protection model makes sealed bids genuinely confidential, and its ZK-based identity layer allows the Haunt Score to accumulate reputation without exposing user identity before settlement.

The system follows a strict separation: funds live on-chain (Midnight), logic executes off-chain (Intent Server + Runner), and trust is anchored in smart contract invariants.

Figure 1 shows the three-layer architecture at a glance. Each layer is examined in detail in the sections that follow.



Core principle: Funds live on *Midnight*. Logic lives *off-chain*. Privacy enforced by *ZK proofs*.

Figure 1. High-level three-layer architecture of GHOST Finance. Layer 3 runs entirely on Midnight Network, whose native zero-knowledge proof system enforces bid confidentiality and pseudonymous reputation without revealing user identity.

Midnight Network: The Settlement Layer

Midnight Network is a data-protection blockchain built as a partner chain of Cardano. It enables smart contracts that can selectively disclose information — preserving confidentiality where needed while remaining publicly verifiable where required. For GHOST Finance, this is not a nice-to-have: the core protocol mechanics *require* a chain where bids can be sealed at the network level, not merely hidden by convention.

The three specific properties GHOST relies on are: **(1) Shielded contract state** — lender and borrower bids are encrypted at submission and only revealed to the matching engine at epoch close. **(2) ZK-based identity** — the Haunt Score is tied to a zero-knowledge identity, allowing reputation to accumulate without linking wallet addresses. **(3) Programmable disclosure** — settlement results (clearing rate, matched amounts) are published publicly; individual bids are not.

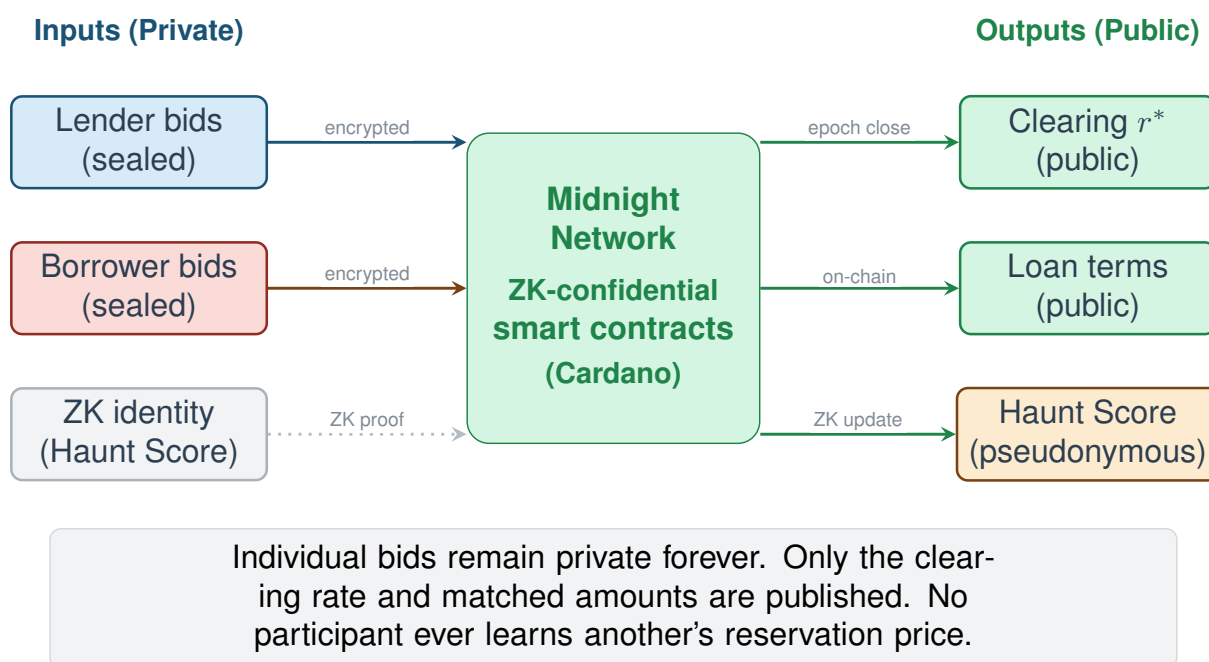


Figure 2. Midnight Network as the settlement layer. Inputs enter sealed via ZK proofs; only aggregate settlement outputs are made public.

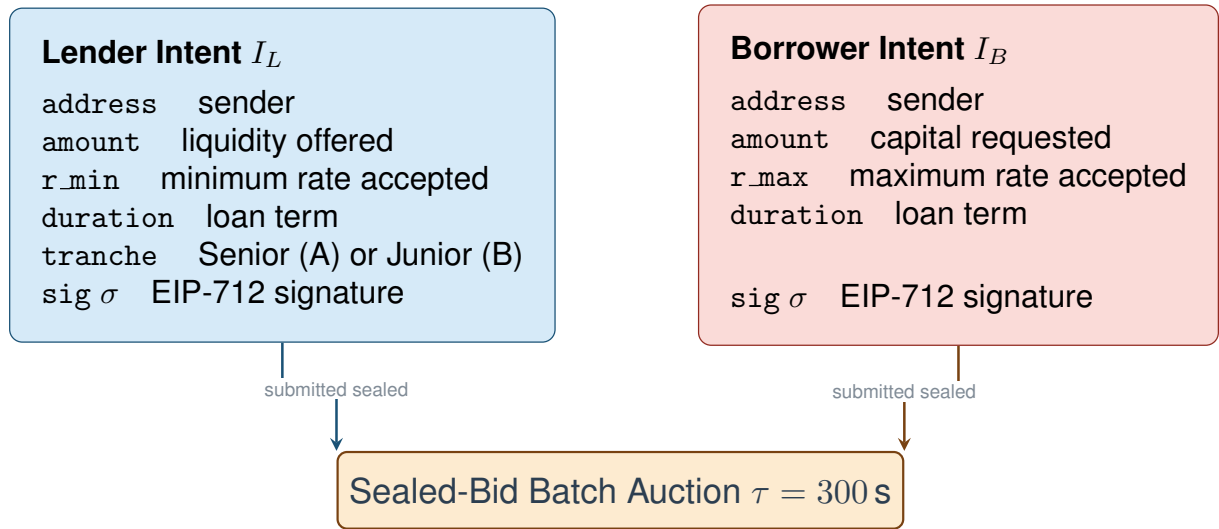
Batch Auction Mechanism

The Problem with Continuous Markets

Existing protocols such as Aave and Compound update interest rates every block using a utilisation curve. This creates two problems: (1) rates are unpredictable, and (2) on-chain order books leak information to MEV searchers who can front-run large deposits.

GHOST replaces the continuous market with a **sealed-bid batch auction** running in discrete epochs of $\tau = 300$ seconds. Participants submit encrypted intents; the clearing engine computes a single clearing rate r^* at epoch close. No bid is visible before settlement.

Intent Data Structures



No participant observes any other bid before epoch close.

Figure 3. Intent data structures for lenders and borrowers. Both are submitted as sealed messages, preventing information leakage before settlement.

Clearing Rate Calculation

After the epoch closes, the matching engine builds cumulative supply $S(r)$ and demand $D(r)$ curves and finds the rate that maximises matched volume:

$$r^* = \arg \max_r \{ \min(S(r), D(r)) \}$$

where

$$S(r) = \sum_{i: r_{\min,i} \leq r} \text{amount}_i, \quad D(r) = \sum_{j: r_{\max,j} \geq r} \text{amount}_j.$$

All matched participants transact at r^* regardless of their reservation price. This is a uniform-price auction: truthful bidding is the dominant strategy, and MEV is structurally eliminated.

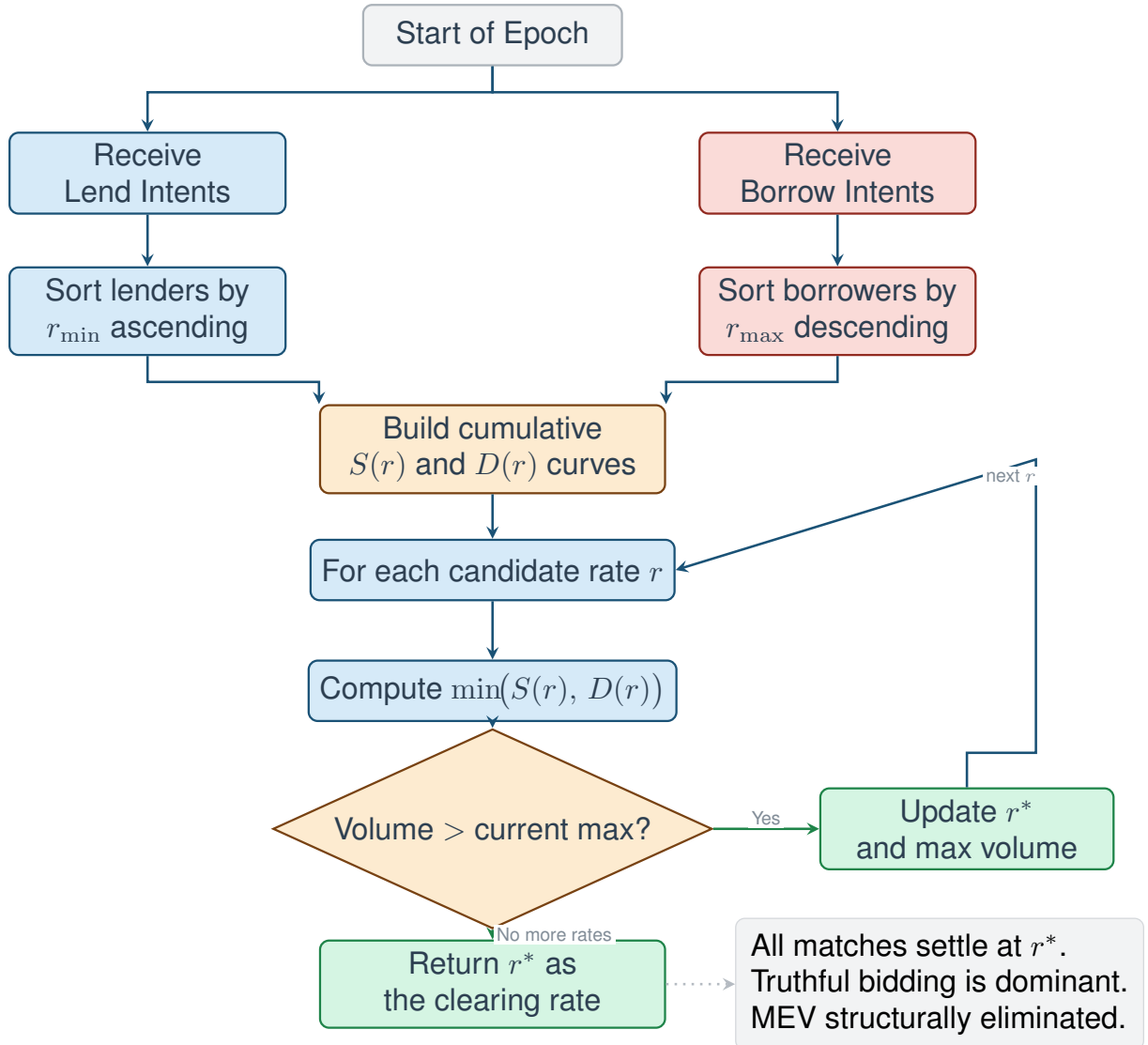


Figure 4. Clearing rate calculation flowchart. The engine iterates over all candidate rates and selects the one maximising matched volume. Every matched participant transacts at the same r^* regardless of their individual reservation price.

Tranched Risk Architecture

Principal Split

Not all capital has the same risk appetite. A DAO treasury requires predictable, protected yield. A yield farmer may accept first-loss risk for a higher return. GHOST splits each loan into two tranches inspired by structured finance.

First priority. Rate = r^* . Protected until loss exceeds 30%.

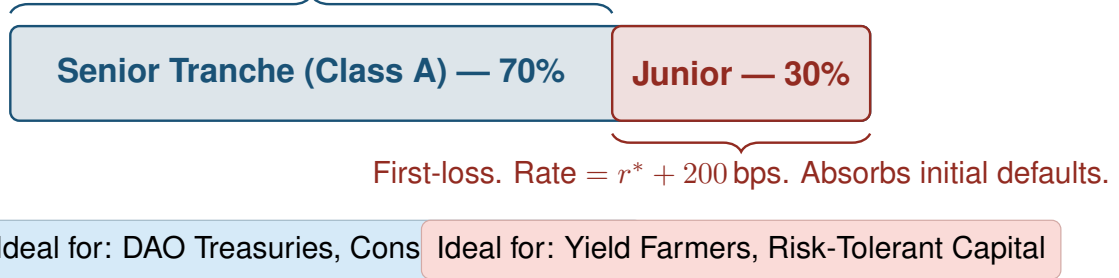


Figure 5. Principal split across tranches. The 70/30 split means Junior absorbs the first 30% of any default loss before Senior is touched.

Waterfall Distribution

When a loan is repaid or liquidated, the recovered amount R flows through the waterfall below. The smart contract enforces that total payouts never exceed R .

$$\text{senior_owed} = P_s \left(1 + r^* \cdot \frac{t}{365} \right) \quad (1)$$

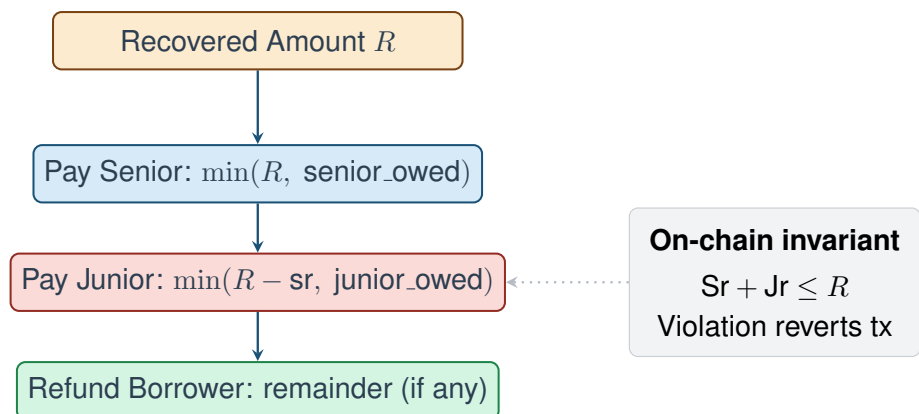
$$\text{junior_owed} = P_j \left(1 + (r^* + s) \cdot \frac{t}{365} \right) \quad (2)$$

$$\text{senior_payout} = \min(R, \text{senior_owed}) \quad (3)$$

$$\text{junior_payout} = \min(R - \text{senior_payout}, \text{junior_owed}) \quad (4)$$

$$\text{borrower_refund} = \max(0, R - \text{senior_payout} - \text{junior_payout}) \quad (5)$$

where $s = 200$ bps, t is elapsed days, and the invariant $\text{senior_payout} + \text{junior_payout} \leq R$ is enforced on-chain.

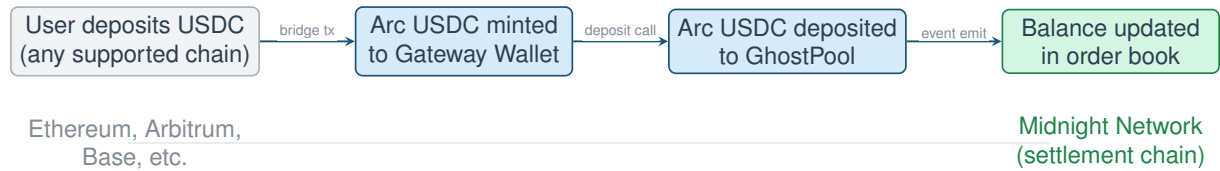


If $R \geq \text{senior_owed}$: Senior fully paid. If $R < \text{senior_owed}$: Junior receives nothing.

Figure 6. Waterfall distribution. Funds cascade Senior → Junior → Borrower. The contract enforces payouts never exceed recovered funds.

Cross-Chain Deposit Flow

GHOST aggregates liquidity across multiple chains into a single order book. Deposits on any supported chain are bridged to **Midnight Network** — the settlement chain — via Arc USDC minting. Once on Midnight, funds are registered in the GhostPool contract and become available to the sealed-bid matching engine.



Capital is fungible across chains. The order book sees a single USDC pool regardless of deposit origin.

Figure 7. Cross-chain deposit flow. Deposits on any chain are normalised to Arc USDC on the settlement chain before entering the order book.

Haunt Score: On-Chain Credit Reputation

Motivation

DeFi protocols require 150%+ collateral from every user, regardless of history. GHOST introduces the **Haunt Score** — an on-chain reputation system that accumulates with good behaviour and reduces the collateral burden over time. Identity remains pseudonymous; only on-chain behaviour is recorded.

Score Accumulation and Collateral Formula

The score evolves as:

$$H(a, t) = H(a, t - 1) + \Delta, \quad \Delta = \begin{cases} +10 & \text{on successful repayment} \\ -H & \text{on default (full reset)} \end{cases}$$

The required collateral ratio decreases linearly from 150% to 110% as the score rises from 0 to the cap of 400:

$$C(H) = 150\% - 40\% \cdot \min\left(\frac{H}{400}, 1\right)$$

Collateral ratio $C(H)$

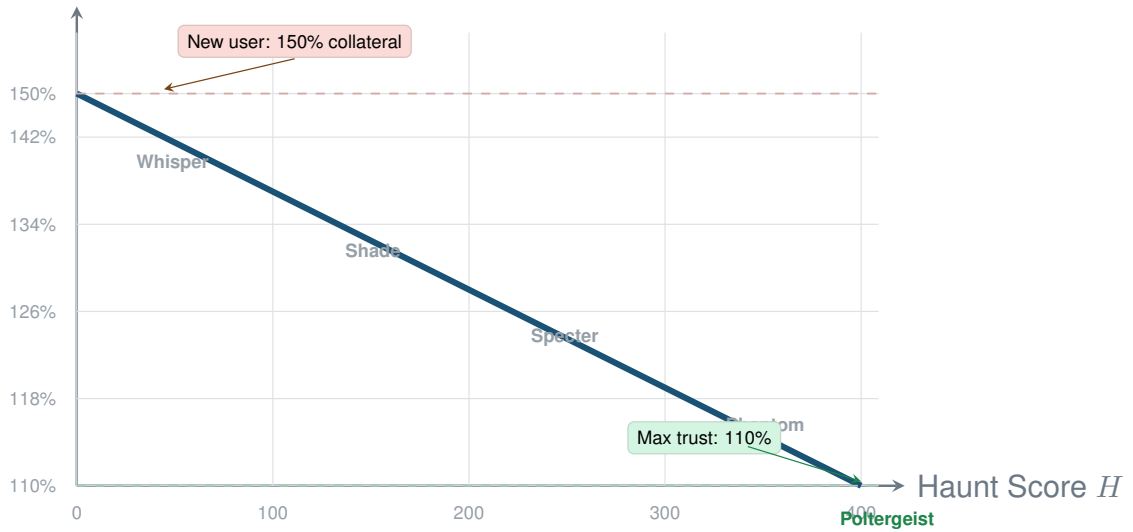


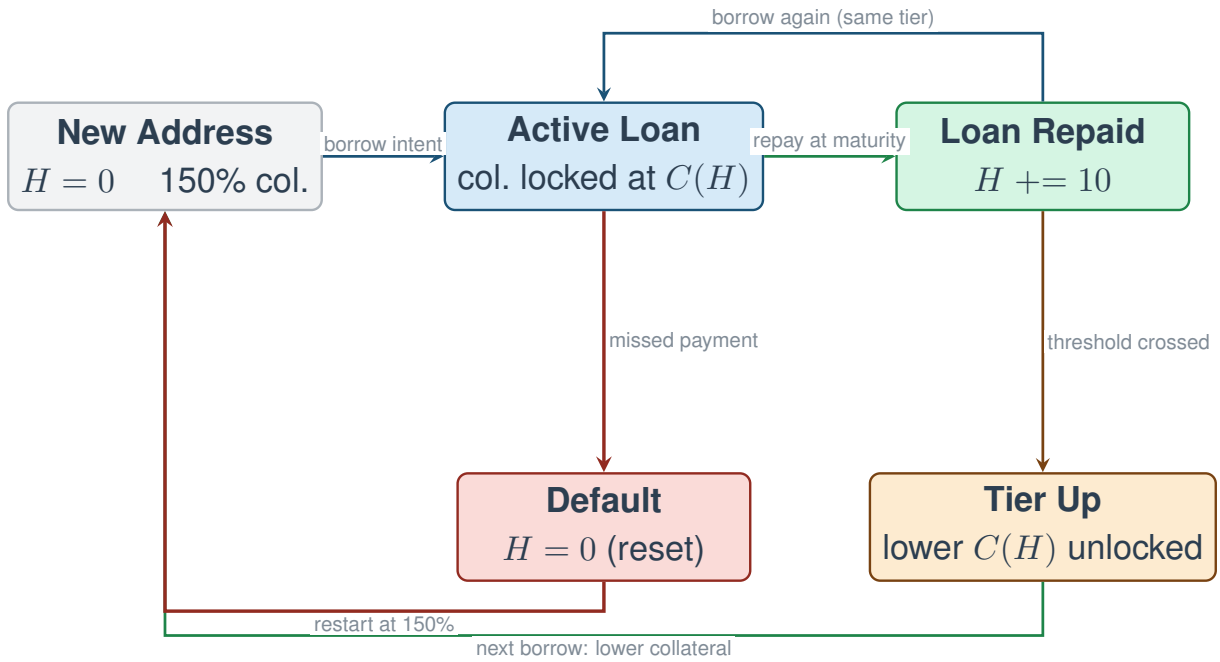
Figure 8. Collateral ratio $C(H)$ as a function of Haunt Score. The linear decrease from 150% to 110% incentivises consistent repayment. A default resets the score to zero.

The Five Manifestation Tiers

Level	Name	Score Range	Collateral	Status
1	Whisper	0 – 99	150%	New user — untested
2	Shade	100 – 199	140%	Emerging history
3	Specter	200 – 299	130%	Established presence
4	Phantom	300 – 399	120%	Strong track record
5	Poltergeist	400+	110%	Maximum trust

Figure 9. The five Haunt Score manifestation tiers. Staking GHOST tokens accelerates accrual: $\text{effective gain} = \text{base gain} \times (1 + \min(\text{staked}/1000, 0.5))$.

Score Lifecycle State Machine



Tier	Score	Collateral
Whisper	$H < 100$	150%
Shade	$100 \leq H < 200$	140%
Specter	$200 \leq H < 300$	130%
Phantom	$300 \leq H < 400$	120%
Poltergeist	$H \geq 400$	110%

Figure 10. Haunt Score lifecycle state machine. Happy path (top row): each repayment adds +10 and may promote the user to a lower-collateral tier. Default (bottom-left) resets the score to zero and restores 150% collateral.

GHOST Tokenomics

Token Distribution

Total fixed supply: **100,000,000 GHOST tokens**.

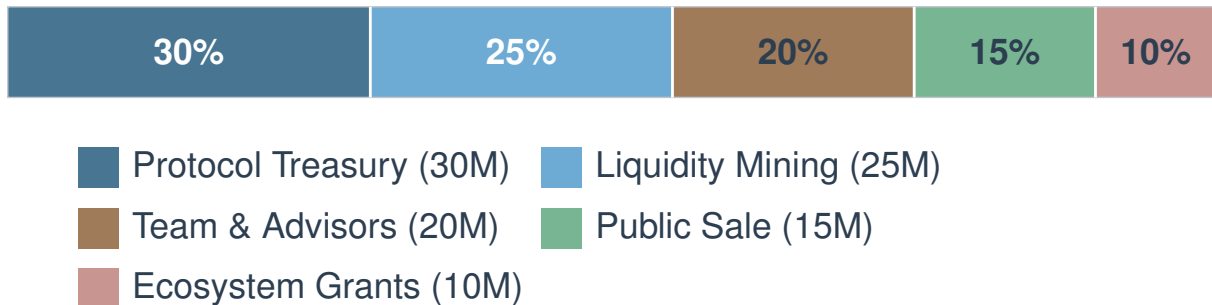


Figure 11. Token distribution across the 100M fixed supply.

Token Utility Map

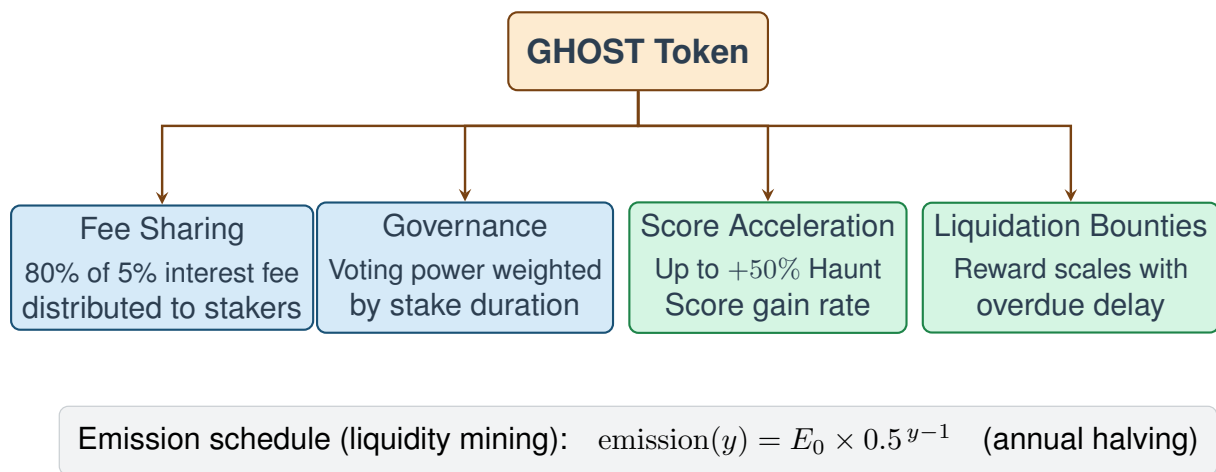


Figure 12. The four utility functions of the GHOST token. Staking provides both economic returns (fee sharing) and protocol benefits (score acceleration).

Protocol Invariants and Security Model

The GhostPool smart contract enforces three mathematical invariants on every state-changing call. A violation causes the transaction to revert.

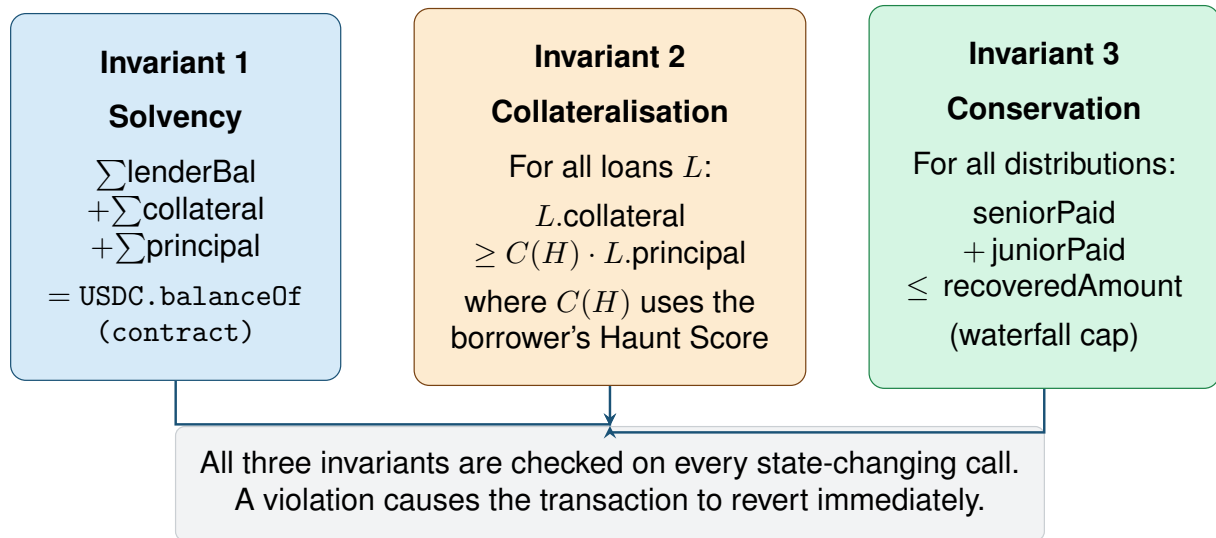


Figure 13. The three on-chain protocol invariants. These cannot be violated by any external call — they are checked at every state transition.

End-to-End Transaction Flow

This final diagram traces a complete lending cycle from intent submission through repayment and score update. It integrates all components described in the preceding sections.

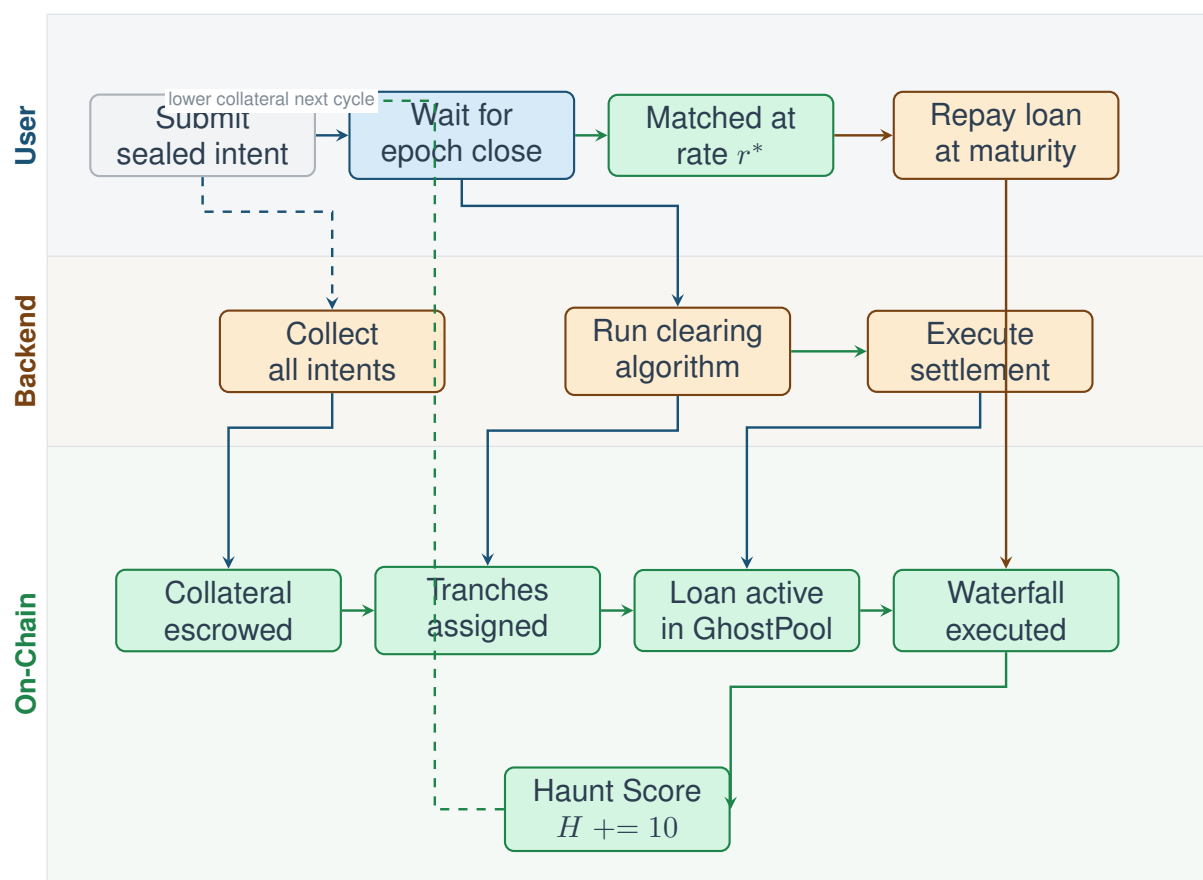


Figure 14. End-to-end transaction lifecycle. The dashed feedback arrow shows how a successful repayment improves the Haunt Score, reducing the collateral required in the next borrowing cycle.